Matching the National Longitudinal Survey of Youth to the Decennial Census: Match Rates and Reasons for Not Matching

by

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Abstract

In this paper, we examine linkages between the National Longitudinal Survey of Youth (NLSY) 1979 and 1997 sample members and the restricted version of the 2020 Decennial Census (DC), focusing on the match rate and the reasons for not matching. Match rates between the NLSYs and the 2020 DC differ by sex and race/ethnicity. Men as well as Black and Hispanic individuals are less likely to match to the DC. Differences in match rate are explained in part by differences in rates of death, in the likelihood of being assigned a linking identifier, and in the likelihood of being enumerated in the DC. This is part of a larger project to compare information about NLSY sample members who have missed several rounds with those who respond regularly. The representativeness of the matched samples affects the capacity of the linked data for evaluating nonresponse bias in the NLSY79 and NLSY97.

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I. Introduction

Numerous studies show that linked data sets are valuable for assessing economic mobility, understanding how employers and schools affect outcomes, and measuring poverty and the effectiveness of the safety net.[1] The linking of data sets can expand the information available about the entities being studied. However, if the linkages are incomplete, the representativeness of the linked data may be diminished (Bond, et al. 2014; Raze, Perales, and Landivar 2026; Silverwood, et al. 2024).[2]

In this paper, we examine linkages between the National Longitudinal Survey of Youth (NLSY) 1979 and 1997 sample members and the restricted version of the 2020 Decennial Census (DC), focusing on the match rate and the reasons for not matching. The records from the two sources are matched at the individual level via the Census Bureau's Protected Identification Key (PIK). We undertook this linking as part of a larger project to compare information about NLSY sample members who have missed several rounds with those who respond regularly. The issue of representativeness of the matched samples is fundamental to our eventual aim: using these data to evaluate nonresponse bias in the NLSY79 and NLSY97.

Though other studies examine the rates of not matching between administrative records and survey data (e.g., Meyer, Mittag, and Goerge 2022; Silverwood, et al. 2024; Miller, Wherry, and Aldana 2022), we believe this study is the first to parse the reason for

[1] For example, Chetty, et al. (2014); Abowd, Kramarz, and Margolis (1999); Card, Heining, and Kline (2013); Aizer, et al. (2016); and Meyer and Mittag (2019).
[2] Raze, et al. (2026) show that weighting techniques can ameliorate the bias from selection into matched data.

not matching. The NLSYs attempt to interview all living sample members at every interview round regardless of where they are living and record the deaths of sample members. These survey policies permit us to assign the reason for not matching for some non-matches, with the residual explanation being that the NLSY sample member cannot be matched to one individual in the DC.

We find that 73 percent of NLSY79 and NLSY97 sample members can be matched to the 2020 DC. Match rates vary substantially by race/ethnicity and sex. Overall, we can assign a reason, other than the residual explanation, for 51 percent and 23 percent of the sample members in the NLSY79 and NLSY97 who could not be matched to the 2020 DC, respectively. For both cohorts, the residual explanation of finding no single individual in the DC with the NLSY individual's PIK is the largest explanation for not matching, both overall and across demographic groups.

II. Previous Literature

Other researchers match samples of the U.S. population to the DC. For example, Colmer, et al. (2024) and Stinson and Weiwu (2026) match samples from 1040 tax files to the 2000 and 2010 DC to retrieve race and ethnicity and thus examine Black and White individuals separately. A team of researchers linked the 1940 DC to several surveys of aging to examine how childhood residential environment affects outcomes over the life course and the accuracy of childhood information collected retrospectively (Lee, et al. 2025; Warren, et al. 2020; Warren, Lee, and Osypuk 2022).

Miller, Wherry, and Aldana (2022) report PIK rates and match rates in their study that links California birth records from 1984 to 1994 to the 2000 DC and 2001–2011 waves of

the American Community Survey (ACS). They link these datasets to identify siblings born in California before and after an expansion in Medi-Cal prenatal coverage for immigrants. The PIK rate for the California birth records from this time period is 97.2 overall, but only about 13 percent of records with a PIK can be matched to the Census Bureau surveys (DC or ACS). A number of reasons may explain why the records do not match to the Census surveys, including mothers or children who leave the United States before the Census surveys take place, children who do not reside with their mothers, children who die before being included in a Census survey, and mothers and children who do not appear in the Census surveys. While Miller, et al. (2022) provide possible reasons for not matching, the available data do not contain information that would be needed to attribute the reason for not matching to the birth records.

Because the NLSY are longitudinal data sets and report reason for non-interview at each round of data collection, the NLSY data identify whether respondents had died before a given round. The NLSY data also identify whether sample members had moved outside of the United States. We use that information—together with DC information on whether respondents were living in group quarters and information on whether NLSY sample members were assigned a PIK—to classify non-matches by reason. In our classification, the residual reason is that no one person in the DC has the same PIK as the NLSY sample member.

III. Data

A. National Longitudinal Surveys of Youth

The NLSYs provide comprehensive information about two cohorts of Americans. The older group, the NLSY79, consists of 9,964 individuals born from 1957 to 1964 who were age 14 to 22 and living in the United States when first interviewed in 1979.[3] The younger group, the NLSY97, consists of 8,984 individuals born from 1980 to 1984 who were age 12 to 17 and living in the United States when first interviewed in 1997.

The NLSY79 has collected 31 interview rounds since the late 1970s. Interviews were conducted annually for 16 rounds, through 1994, and biennially hence. The NLSY79 sample members were ages 60 to 68 when the most recent interview was conducted from 2024 to 2025. The NLSY97 has conducted 21 interviews since the survey began in the late 1990s. Similar to the NLSY79, interviews of NLSY97 sample members were collected annually for the first 15 rounds, through 2011, and biennially hence. The 22nd round of interviewing began in March 2026, when the sample members were ages 42 to 47.

The primary focus of the surveys is labor force and employment behavior. However, they take a broad approach, attempting to capture information about domains of life that may impact jobs and earnings. Thus, in addition to asking about employment topics including occupation, working hours, and compensation, the surveys ask about education, health, marriage, children, wealth, criminal behavior, participation in government programs, cognitive skills, attitudes, and expectations.

[3] The sample size of the NLSY79 was originally 12,686. Two subsamples were dropped later. The military oversample was discontinued after Round 5 of data collection in 1984. The poor-white oversample was discontinued after Round 11 in 1990.

The NLSYs are remarkable for their high response rates. Figure 1 shows response rates and retention rates by round for the NLSY79 and NLSY97.[4] Over 40 years after NLSY79 began, 64.4 (74.4) percent of (living) sample members participated in the 30th round of data collection in 2022. In the 21st round of NLSY97 data collection in 2025, 73.1 (75.8) percent of (living) NLSY97 sample members participated in the survey. Survey participation rates vary across demographic groups. Table 1 shows retention rates in the most recent seven rounds overall and by sex and race/ethnicity for both NLSYs. In both surveys, men are less likely to participate than women. In addition, Black men and women are more likely to participate in recent rounds than their counterparts who are Hispanic or non-Black, non-Hispanic.

The NLSYs attempt to interview sample members regardless of where they are living, including those residing in institutions such as nursing homes or jails and prisons and those who live outside of the United States. In the most recent rounds, approximately 3 percent of the sample members for each cohort were unlocatable. Deaths of NLSY sample members are recorded as part of survey operations.

For this project, the full samples (excluding subsamples dropped from the NLSY79) were available for matching to the DC. In accordance with the Privacy Act of 1974, the PIKs for the NLSY can be used only by the Bureau of Labor Statistics (BLS) to evaluate data quality. For example, BLS could use NLSY PIKs to match NLSY data to other data available in the Federal Statistical Research Data Centers (FSRDC) to examine other facets of data

[4] Retention rate is defined as the percentage of sample members who participate in a survey round. In contrast, response rate is the percentage of sample members who participate in a survey round out of those sample members not known to be deceased.

quality, such as the accuracy of reports on program participation or on arrest and incarceration. However, for NLSY PIKs and any matched data created using these PIKs to be used by researchers outside BLS, the NLSY sample members would need to consent for their data to be linked and used.

B. Decennial Census

The goal of the DC is to count every individual in the United States exactly once and in their correct location. In the 2020 DC, self-response could be reported by web, phone, or mail. For households where responses were not received, Census takers visited or called to collect the information. Procedures were implemented to count individuals who lived in group quarters or in transitory situations.

For our analysis of NLSY nonresponse, linkage to the DC provides information on geography and household structure in 2020 for NLSY sample members, regardless of NLSY survey participation. In other research, linkage to the DC is valuable to identify family members and demographic characteristics (e.g., Colmer et al. 2024; Miller et al. 2022).

C. Match Procedures

In 2022, using information provided by BLS about NLSY sample members, including name, date of birth, and addresses, the Census Bureau assigned a PIK to each NLSY sample member using their Person Identification Validation System (PVS).[5] The Census Bureau's PVS matched the NLSY data to a master reference file. Individuals who matched

[5] BLS provided up to three addresses to the Census Bureau to be used in the PVS process. The addresses were the most recent, second most recent, and third most recent available. These addresses were of different vintages depending on how recently BLS had contact with the respondent.

were given a PIK, which acts as an anonymized identifier that can be used to link individuals across datasets (Wagner and Layne 2014; Massey and O'Hara 2014).

All NLSY records that were missing a PIK were dropped from the matched samples. In a very small number of cases, two NLSY79 or NLSY97 sample members had the same PIK. In these cases, we retained one sample member and dropped the other. If possible, demographic information was used to select the sample member who was retained; otherwise, selection was random.

Similarly for the DC, records that are missing a PIK were dropped from the matched samples. When two or more records shared the same PIK, we selected the best match using the PVS variables (which capture match quality along multiple dimensions) and dropped the duplicates. As shown in Table 2, for the 2000 and 2010 DCs, 88 percent of person records have an unduplicated PIK. For the 2020 DC, the rate of records with an unduplicated PIK is 82 percent (Brown, et al. 2023).

Raze, et al. (2026) and references therein show that inability to match to a PIK is nonrandom. Using ACS and Current Population Survey Annual Social and Economic Supplement (CPS ASEC) data for the years 2005 to 2022, Raze, et al. (2026) document that young children, young adults, minorities, immigrants, recent movers, renters, and individuals with less education are less likely to be matched to a PIK and suggest possible explanations for the lower match rates. During these years, PIK rates for the ACS ranged

from 88 to 93 percent. After 2006, yearly PIK rates for the CPS ASEC ranged from 84 to 89 percent.[6]

IV. Match Rates and Reasons for Not Matching

After data preparations, NLSY sample members with PIKs can be matched by PIK to the 2020 DC files. Table 3 shows the match rate for the NLSY79 and the NLSY97 to the 2020 DC.[7] For both NLSYs, the overall match rate to the 2020 DC is 73 percent. Match rates vary by sex and race/ethnicity in comparable ways for the NLSY79 and the NLSY97. About 80 percent of non-Black, non-Hispanic individuals from the NLSYs match to the 2020 DC, but the match rates are lower for Black and Hispanic individuals.[8] Black individuals in the NLSY97 are somewhat less likely to match to the 2020 DC than are Black individuals in the NLSY79 (64 percent versus 67 percent). Women in the NLSYs are more likely to match to the 2020 DC than men; 78 (77) percent of NLSY79 (NLSY97) women match, compared to 68 (70) percent of men.

Table 4 explores some possible reasons for NLSY sample members not matching to the 2020 DC. We construct categories explaining the reason for not matching that are mutually exclusive and exhaustive. For some cases, multiple reasons explain the non-match. To prevent such individuals from being counted more than once, we establish

[6] Beginning in 2006, CPS sample members must opt out of linkage to administrative records, as opposed to opting in.
[7] Corresponding tables were constructed for the 2000 and 2010 matched files but not released from the FSRDC to minimize disclosure risk. We chose to release the 2020 match rates because of their recency.
[8] Here and subsequently for brevity, Black, non-Hispanic individuals in NLSYs are described as Black individuals.

priority among reasons for not matching. As such, the percentages in Table 4 sum to 100 across the five categories for the NLSY as a whole or for any demographic subgroup.[9]

First, individuals who are known to be deceased and do not match to the DC are coded as such.[10] For the NLSY79, the death of sample members before 2020 explains 41 percent of non-matches to the 2020 DC.[11] As with the patterns for non-matches in general, deaths vary by sex and race/ethnicity. Men are 4 percentage points more likely to have died than women. Black individuals are more likely to have died than Hispanic and non-Black, non-Hispanic individuals.[12] For the NLSY97, a smaller percentage of the sample had died. This is expected for a younger cohort, but already the differences by race/ethnicity and sex are emerging, with men and Black individuals more likely to be non-matches because they have died. Deaths explain 10 percent of non-matches to the DC. Match rates excluding the

[9] To comply with Census Bureau disclosure avoidance requirements for unweighted summary statistics and the counts that underlie unweighted proportions, we opted to apply noise injection to the counts before computing rates (FSRDC Disclosure Avoidance Methods: A Handbook for Researchers, Version 5.0, October 10, 2024). Because Table 4 shows rates by sex and by race/ethnicity, we opted to apply noise injection to the subgroup counts (by sex and race/ethnicity) and then summed the noise-injected subgroup counts to get overall counts. We applied a raking procedure to make the subgroup counts consistent with the overall counts.

[10] Information about deaths comes from the NLSY 'reason for non-interview' variable. We use this variable for 2020 for the NLSY79 and 2019 for the NLSY97. The timing of death information does not align with Census Day (April 1, 2020). For example, an NLSY79 sample member who died in September 2020 may have participated in the DC in April 2020. If this sample member matched to the DC, we would code this sample member as a match.

[11] The numbers in Table 4 are the shares of a cohort that match to the DC or do not match for a given reason. When we report the share of non-matches represented by a particular reason, we divide the relevant number by the percent of non-matches. For example, for the NLSY79, 27 percent (100 – 73) of the cohort did not match to the DC and 11 percent of the cohort did not match due to being deceased. Therefore, the death of sample members represents 41 percent (11/27) of non-matches.

[12] See Brummet, Rothstein, and Gardecki (2026) for more information on mortality in the NLSY79 by age 58. Weighted descriptive statistics show that within race, men have higher mortality rates than women. The greatest differential is for Black, non-Hispanic individuals: Black men are almost 6 percentage points more likely to be deceased by age 58 than Black women.

deceased from the denominator are 82 percent for the NLSY79 and 75 percent for the NLSY97.

Then, among the not known dead, those who do not match to the DC and are identified to be living outside of the United States or in group quarters are coded. Location information is available for NLSY sample members who participate in the relevant round. Indicators for living outside of the United States (50 states) are available in 2020 for NLSY79 and in 2019 for NLSY97. Individuals living in group quarters are part of the 2020 DC and may be assigned a PIK. Thus, we have linked records for some NLSY sample members living in group quarters at the 2020 DC. But because the DC records for individuals who are living in group quarters lack many of the household variables, such as those describing household structure that we use in our nonresponse analysis, we drop them from our matched samples. Because the number of NLSY sample members who do not match to the 2020 DC because they are living abroad or in group quarters is small, we combine those categories.

A small percentage of NLSY79 non-matches can be explained by either living outside the U.S. or in group quarters at the time of the 2020 DC (5 percent). For the NLSY97 also, living abroad or in group quarters explains a small percentage of the non-matches for the cohort (8 percent). For both cohorts, the rate of non-match for this reason is higher for men than women.

Next, for the remaining unmatched NLSY sample members, those whose NLSY record does not have a PIK or has a duplicate PIK constitute the next category. This is a small percentage of all sample members (1.5 percent in the NLSY79 and 1.4 percent in the

NLSY97), indicating that the PVS process performed well for the NLSY samples. However, for Hispanic individuals in the NLSYs the proportion who do not have a PIK or have a duplicate PIK is higher (4.6 percent in the NLSY79 and 3.6 percent in the NLSY97), like findings from an assessment of the PVS system using 2009 ACS data (Mulrow et al., 2011).

Finally, the residual group is composed of those NLSY sample members with a PIK for whom a match cannot be found in the 2020 DC. We interpret this category as non-matching due to the DC record not having a PIK or having a duplicate PIK; this category also includes individuals who were in the DC universe but were not enumerated.

For both the NLSY79 and NLSY97, the largest reason for not matching—at 13 and 21 percent of sample members (or 48 percent and 78 percent of sample members who are not matched), respectively—is that no person in the 2020 DC has the individual's PIK. Like match rates, the rates of no person in the 2020 DC having the individual's PIK vary by sex and race/ethnicity, with non-matching rates due to lack of a unique PIK in the 2020 DC higher for men and for Black and Hispanic individuals.

Having no person in the 2020 DC with the corresponding PIK occurs because either the individual could not be assigned a PIK or the individual was not enumerated in the 2020 DC. PIK rates for the 2020 DC vary in the same ways as described above. While the overall PIK rate was 83.6 percent, it was lower for men (82.9 percent), Hispanic people (74.1 percent), and those who report their race as Black alone (79.4 percent) (Brown, et al. 2023).[13] In addition, Census Bureau analyses showed that Black and Hispanic individuals

[13] The PIK rate of 83.6 percent differs from 81.7 percent presented in Table 1 of this paper. Both rates come from Brown, et al. (2023). The latter number removes duplicated PIKs from the numerator, but the former number includes duplicate PIKs in the numerator.

were less likely to be enumerated in the DC. In the 2020 DC, undercounts and overcounts varied by subgroup. Hispanic people, Black people, and American Indians on reservations were undercounted, while White and Asian people were overcounted (National Academies of Sciences, Engineering, and Medicine 2023).

V. Conclusions

The PVS process is quite successful for the NLSY samples. Only 1.5 percent of the NLSY79 and 1.4 percent of the NLSY97 could not be matched to the 2020 DC due to not having a PIK or having a duplicate PIK.

We find that 73 percent of NLSY79 and NLSY97 sample members can be matched to the DC. In our classification of non-matches, one explanation is that NLSY sample members have died prior to 2020. This reason accounts for 41 percent of non-matches for the NLSY79 and 10 percent of non-matches for the NLSY97. For both cohorts, the residual explanation of finding no single individual in the DC with the NLSY individual's PIK is the largest explanation for not matching. Moreover, match rates differ by sex and race/ethnicity. Most consequential for our project on nonresponse, men in the NLSY79 and NLSY97 samples match to the DC at lower rates than women by differences of 10 percentage points and 7 percentage points, respectively. This pattern is consistent with the lower PIK rates of men in the 2020 DC (Brown, et al. 2023).

In our related work, we will be using the matched NLSY-DC data to estimate the effect of NLSY nonresponse. In this paper, we have shown that matching the NLSY and DC is not random; that is, a match is more likely for some demographic groups than others. As a result, restricting to the NLSY records that match to the DC introduces some selection

into the resulting sample. Men have lower match rates than women, and men are less likely to participate in the NLSYs. This pattern means that the matched NLSY-DC data may be less useful for examining the effect of nonresponse on estimates of key labor market outcomes such as labor supply or earnings. The patterns of differential match rates will need to be considered when we assess the effect of NLSY nonresponse using the matched NLSY-DC data.

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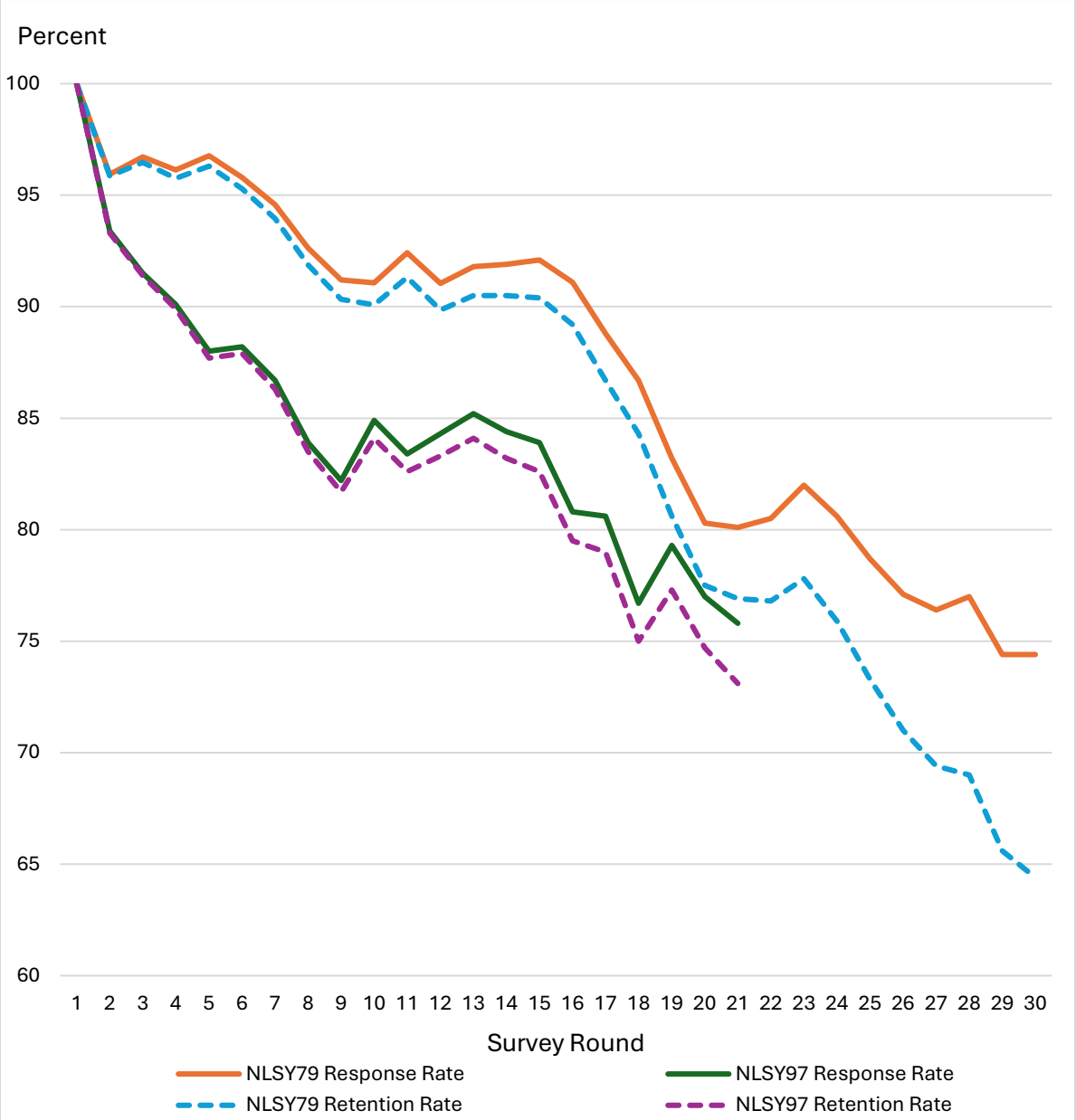
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Figure 1. Response Rates and Retention Rates, By Round, NLSY79 and NLSY97



Note: The non-dropped NLSY79 sample, 9,964 individuals, was used for the calculations in this figure. See footnote 3 for more information on the dropped subsamples. NLSY97 sample size: 8,984. Retention rate is defined as the percentage of sample members who participate in a survey round. In contrast, response rate is the percentage of sample members who participate in a survey round out of those sample members not known to be deceased. NLSY79 was conducted annually through Round 16 and biennially hence. NLSY97 was conducted annually through Round 15 and biennially hence.

Source: NLSY79, NLSY97, and authors' calculations for NLSY79 Rounds 1 to 11 for the non-dropped NLSY79 sample. From nlsinfo.org otherwise.

Table 1. NLSY Retention Rates in Recent Rounds

A. NLSY79

		Females			Males		
Round of Data Collection/Year	All	Black, Non-Hispanic	Hispanic	Non-Black, Non-Hispanic	Black, Non-Hispanic	Hispanic	Non-Black, Non-Hispanic
24/2010	75.92	82.13	76.42	77.86	75.13	70.94	72.60
25/2012	73.26	80.43	74.80	74.68	72.24	68.71	69.46
26/2014	70.96	79.62	71.63	71.94	70.34	66.26	66.84
27/2016	69.37	77.45	69.80	70.77	68.70	64.22	65.49
28/2018	69.03	76.44	71.63	70.09	68.24	65.24	64.58
29/2020	65.59	73.80	69.49	67.95	62.99	59.84	60.72
30/2022	64.36	71.29	69.03	67.19	61.75	57.80	59.81

B. NLSY97

		Females			Males		
Round of Data Collection/Year	All	Black, Non-Hispanic	Hispanic	Non-Black, Non-Hispanic	Black, Non-Hispanic	Hispanic	Non-Black, Non-Hispanic
14/2011	82.62	89.45	86.36	80.02	84.94	82.50	79.15
15/2013	79.47	88.51	83.77	77.98	80.41	77.48	75.18
16/2015	79.06	88.16	82.79	77.75	80.75	77.28	74.26
17/2017	74.96	84.04	80.08	76.07	73.57	69.09	70.49
18/2019	77.24	85.25	82.25	77.53	76.08	71.95	73.60
19/2021	74.65	83.79	80.30	74.38	75.71	69.19	70.29
20/2023	73.07	82.59	78.25	73.31	73.56	68.17	67.97

Source: NLSY79, NLSY97, and authors' calculations.

Table 2. PIK Assignment in Decennial Census, 2000, 2010, and 2020

Year	Total Person Records	Unduplicated PIKs	Percent with Unduplicated PIK
2000	281,421,906	247,117,007	87.81
2010	308,745,538	273,041,579	88.44
2020	331,400,000	270,700,000	81.68

Source: Massey, et al. (2018), Brown, et al. (2023).

Table 3. Match Rates between NLSY and 2020 Decennial Census

	NLSY79	NLSY97
All	73	73
Sex		
Female	78	77
Male	68	70
Race/ethnicity		
Black, non-Hispanic	67	64
Hispanic	66	67
Non-Black, non-Hispanic	80	80

Note: Noise injection was applied to the counts before rates were computed. Rates are rounded to two significant digits.

Source: NLSY79, NLSY97, 2020 Census, and authors' calculations. DRB release number CBDRB-FY26-063.

Table 4. Percent of NLSY and 2020 Decennial Census Matched and Not Matched, by Reason

A. NLSY79

		Reason for Not Matching			
	Matched	Deceased	Living Outside of U.S. or in GQ	NLSY No PIK or Duplicate PIK	2020 DC No PIK or Duplicate PIK
All	73	11	1.3	1.5	13
Sex					
Female	78	9.0	0.6	1.2	11
Male	68	13	2.0	1.8	15
Race/ethnicity					
Black, non-Hispanic	67	14	1.6	1.2	17
Hispanic	66	11	2.5	4.6	16
Non-Black, non-Hispanic	80	9.5	0.6	0.5	9.4

B. NLSY97

		Reason for Not Matching			
	Matched	Deceased	Living Outside of U.S. or in GQ	NLSY No PIK or Duplicate PIK	2020 DC No PIK or Duplicate PIK
All	73	2.6	2.1	1.4	21
Sex					
Female	77	1.7	0.8	1.1	19
Male	70	3.5	3.4	1.7	22
Race/ethnicity					
Black, non-Hispanic	64	3.4	2.9	1.2	29
Hispanic	67	2.5	1.5	3.6	25
Non-Black, non-Hispanic	80	2.3	1.9	0.6	15

Note: GQ—Group Quarters. Noise injection was applied to the counts before rates were computed. Rates are rounded to two significant digits. Because of rounding, rates in a row may not add to 100.

Source: NLSY79, NLSY97, 2020 Census, and authors' calculations. DRB release number CBDRB-FY26-063.